



UMS
UNIVERSITI MALAYSIA SABAH

KP24603 ENTERPRISE NETWORKS AND WIRELESS TECHNOLOGIES
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ASSIGNMENT 1
DETERMINATION OF THE SHORTEST PATH IN THE UMS MAIN CAMPUS
USING THE DIJKSTRA ALGORITHM

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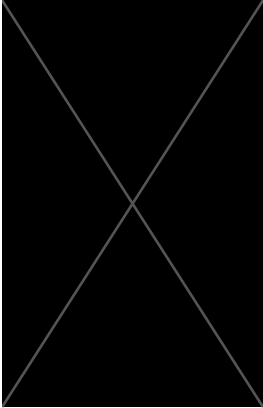
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TABLE OF CONTENT

1.0 Abstract.....	2
2.0 Introduction.....	3
3.0 Methodology.....	5
4.0 Results.....	10
5.0 Conclusion.....	18
6.0 References.....	19

1.0 Abstract

The UMS main campus is large, making navigating difficult for students, staff, and visitors. This study uses Dijkstra's algorithm, a well-known graph-based optimisation approach, to determine the shortest paths across campus, hence increasing navigation efficiency. The study requires detailed data collecting, such as mapping the campus plan, identifying critical places (nodes), and calculating distances (weights) between them. This data is used to create a weighted graph that depicts the UMS campus. Dijkstra's algorithm will be used to find the shortest pathways between different origin-destination pairs. The study's goal is to improve campus accessibility and efficiency, allowing users to easily navigate the campus. The expected results include a full weighted graph of the UMS campus, shortest path calculations, and an evaluation of the algorithm's effectiveness in this context. These findings will be critical in building improved navigation systems, wayfinding applications, and signage solutions for the campus. Furthermore, this study emphasises the practical implementation of graph-based optimisation algorithms in real-world scenarios, demonstrating their efficacy in complicated environments such as university campuses. By tackling these difficulties, the study helps to improve campus navigation.

2.0 Introduction

The UMS main campus is spread over a vast area and is complex in nature, posing problems in finding ways around for the students, staff as well as the visitors. Scheduling the shortest time between specific destinations within the campus is essential in terms of time consumption and energy saving as well as providing better users' experience (Balasubramaniana et al. , 2016; Huang et al. , 2018). Algorithms for solving the shortest path problem on a weighted graph is a well known algorithm; Dijkstra algorithm (Dijkstra, 1959). Dijkstra algorithm is one of the popular graph based optimization algorithms which is used in various researches related to transportation, logistics and communication networks (Goyal et al. , 2021; Rani and Sikka, 2020). If the UMS main campus can be represented as a weighted graph in which the nodes are the main locations in the campus and the edges are the paths that connect these locations the Dijkstra algorithm can be applied to the calculation of the shortest path between two points of the campus.

The present study intends to exploit the Dijkstra algorithm to pinpoint the shortest distances between buildings in the main campus of UMS. This research is particularly important because it can help to improve the navigation for the campus community that include students, staff and visitors (Li et al. , 2020). The results of the present study can inform the implementation of path finding systems on campus either through navigation systems applications or static signage to aid users in the identification of the shortest walks on campus for increased accessibility and enhanced user experience. The research process will start with data collection whereby the data collection process will entail mapping the campus environments in the campus and identifying key locations and calculating distances or weights between the various points on the map. The obtained data will then be used to create the weighted graph to represent the specific main campus of UMS which will be used by the Dijkstra algorithm.

The application of the Dijkstra algorithm in this case can ensure a positive influence on the navigational experience of the campus community. Moreover, the use of the Dijkstra algorithm in a highly specific situation of the UMS main campus may be helpful to reveal the potentiality of graph-based optimization methods in the real world environment. The study will also seek to answer questions regarding the way the algorithm copes with these particular

characteristics and restrictions specific to the campus environment, for example, the existence of buildings and walkways, and possible obstacles, which may help to better understand how effective this algorithm is in other such complex environments like a campus. The practical and academic importance of this research include becoming a more practical perspective, results of this study may easily be incorporated into university navigation systems and mobile applications so that pathfinding can become as smooth and problem-free as possible. This can enhance the ease of navigation and the access ability in the UMS main campus as people get to various destinations they need.

On the theoretical side, the conducted work might be useful as a reference for researchers and practitioners who are concerned with applying graph-based optimization algorithms to navigate in campus or urban spaces. The explanation of the methodological procedure used for this paper, the process of data collection, and the performance results of the implementation of the Dijkstra algorithm in the UMS main campus can serve as references to further studies on the application of advanced algorithms and technologies for easier wayfinding and accessibility in large campus environments. This study is expected to achieve some goals include, producing a weighted graph representation of the main campus of the UMS, development of the Dijkstra algorithm applied to the candidate graph, evaluation of the performance of the Dijkstra algorithm in this specific situation. The research team will also investigate the possible implementation of the findings in campus navigation systems and potential future research directions in this area.

3.0 Methodology

Research Area

The scope of this research is the campus of the University Malaysia Sabah (UMS), located in Kota Kinabalu, Sabah. The campus is large and has a range of infrastructure and amenities that make it perfect for testing and applying the Dijkstra algorithm to find the shortest paths. The academic buildings, administrative buildings, residential areas, and recreational facilities are all part of the campus's multiple components. To ensure the study's applicability and relevance, this research focuses on the developed regions where the majority of staff and student activities occur. The particular field of study includes residential areas like staff quarters and dorms for students, academic complexes like lecture halls, research centres, and faculties, administrative buildings housing the university administration, and recreational spaces like parks, sports complexes, and student activity centres. Together, these components cover a significant subset of the total campus area, providing a comprehensive setting for the study.

Within this defined area, the Dijkstra algorithm will be implemented, making use of the established network of paths and amenities. The algorithm will be used to identify the best pathways between different points of interest on the campus map, improving campus management and navigation. The main infrastructure and facilities are grouped in the northern section of the UMS campus, which is shown in a comprehensive topographic diagram, while the southern section is set aside for future growth and development. The meticulous design guarantees that the research precisely mirrors actual situations, offering significant perspectives and useful recommendations. The Dijkstra algorithm may be effectively applied on the UMS campus, showcasing its ability to optimise routes in a complicated environment. This research not only contributes to academic knowledge but also has significant practical benefits for improving campus navigation and operational efficiency.

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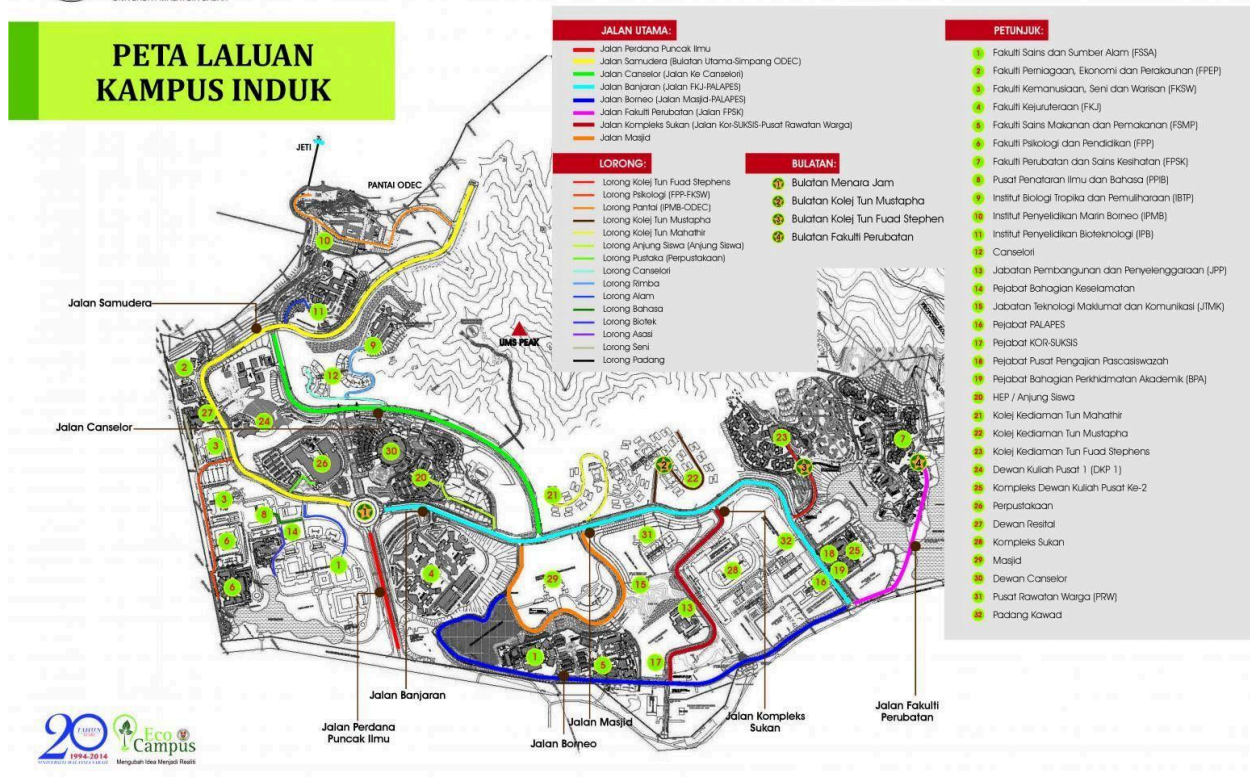


Figure 1 — The main campus of University Malaysia Sabah (UMS)

Structure of the graph

Notation:

G :Adjacency matrix representing the graph.

$G(i, j)$ weight of the edge from node i to j

V_s : Source node ,starting point or origin node

d : Distance vector where $d[i]$ is the shortest known distance from the source node s to node i

π :Predecessor vector where $\pi[i]$ is the predecessor of node i on the shortest path from the source node s to i

Q : Logical vector that tracks unvisited nodes. $Q(i) = true$ for all nodes

Dijkstra's algorithm aims to determine the shortest path in a weighted graph with non-negative edge weights between a source node, V_s to all other nodes. this research using Matlab to find out the shortest path,this algorithm can be implemented using an adjacency matrix G ,where $G(i, j)$ represents the weight of the edge from node i to node j . The distance vector d is initialised at the beginning of the method by setting $d[s] = 0$ for the source node and $d[v] = \infty$ for all other nodes v . To evaluate the shortest path tree,a predecessor vector $\pi[i]$ indicate the predecessor of node i . A logical vector Q is used to track unvisited nodes,with all nodes initially marked as unvisited $Q(i) = true$.

The procedure then identifies the node u from the collection of unvisited nodes Q that has the least distance $d[u]$ while there are unvisited nodes. After then, this node u is designated as visited by setting $Q[u]=false$. The algorithm determines if each neighbour, v of the chosen node, u (i.e,nodes where $G(u, v)>0$), the algorithm determines whether the neighbour v has not yet been visited. Assuming v remain unexplored,the method determines an approximate distance by u as $d[u] + G(u, v)$. If this tentative distance is smaller than the current known distance $d[v]$, the algorithm updates $d[v]$ to this new distance and set $\pi[v]$ to u , indicating that the shortest path to v now goes to u .

The procedure keeps going until every node is visited or the distance between the unvisited nodes is infinity, meaning that the nodes that are left cannot be reached from the source node. The final output is the distance vector d ,which includes the shortest path from the source node V_s to each node, as well as the predecessor vector π which can be used to reassemble the shortest paths.

Configuration of graph

The configuration of the graph involves specifying the total number of nodes present in the UMS Main Campus. There are 15 nodes also known as vertices identified in the UMS Main Campus map. These include 1 main gate and 14 building establishments. Table 1 provides an analysis on the nodes of the graph along with detailed explanation for each one.

Table 1- Nodes of the graph

Nodes	Interrelation - Illustration of the nodes
1	Main Gate
2	Fakulti Sains dan Sumber Alam
3	PPIB
4	Library
5	DKP Lama
6	Dewan Resital
7	Fakulti Perniagaan, Ekonomi dan Perakaunan
8	Dewan Canselor
9	Fakulti Kejuruteraan
10	Kolej Kediaman Tun Mustapha
11	Kolej Kediaman Tun Fuad
12	Kolej Kediaman Tun Pengiran Ahmad Raffae
13	Padang Kawad / Kompleks Sukan
14	DKP Baru
15	Fakulti Perubatan dan Sains Kesihatan

The weights are calculated based on the distance a person travels from one building to another. These distances that are measured are not the direct lines connecting the nodes of the graph but rather the actual path in the campus. Table 2 provides a detailed explanation on the total distances representing the edges of the graph. Note that the graph is non-directional which means that the movement of nodes can occur in both directions.

Weights (Distance in metres)

Edges	Weights	Edges	Weights
1-2	900	5-6	260
1-3	850	5-7	400
1-8	650	6-7	260
1-9	900	8-9	600
2-3	400	10-11	900
2-4	500	11-12	1200
3-4	450	12-13	600
3-5	850	12-14	950
4-5	700	13-14	300
4-8	300	14-15	900

Development of Dijkstra algorithm

The Dijkstra algorithm was developed and implemented in the matlab programming environment to calculate the shortest path. Based on the Dijkstra algorithm equation that is implemented as follows ; $[cost, route] = dijkstra (G, Vs, Vd)$. Cost refers to the minimum cost which is the total length of the shortest route while route is the sequence of nodes defining the resulting path. The G refers to the graph that is being analysed whereas Vs is the source node which is the starting point and Vd is the destination node which is the end point.

The Dijkstra algorithm efficiently navigates all possible paths in the University Malaysia Sabah campus graphs that emphasises on the complex paths that shows the effectiveness in finding the shortest path.

4.0 Results

Table 3 below shows the results of all Dijkstra algorithm applications for finding the shortest paths on the main campus of Universiti Malaysia Sabah. The table also shows the source node and destination node which are the starting point and end point from which distances are calculated. Min cost which is the distances of the path are also included. Then routes, which refers to the sequence of nodes that connects starting point to end point.

Table 3- Results of the Dijkstra algorithms

α/α	Source node	Destination node	Min cost	Route	α/α	Source node	Destination node	Min cost	Route
	Vs	Vd	cost	rute		Vs	Vd	cost	rute
1	1	4	950	1-8-4	42	5	10	2600	5-4-8-9-10
2	1	5	1650	1-8-4-5	43	5	11	3500	5-4-8-9-10-11
3	1	6	1910	1-8-4-5-6	44	5	12	4700	5-4-8-9-10-11-12
4	1	7	2050	1-8-4-5-7	45	5	13	5300	5-4-8-9-10-11-12-13
5	1	10	1900	1-9-10	46	5	14	5600	5-4-8-9-10-11-12-13-14
6	1	11	2800	1-9-10-11	47	5	15	6500	5-4-8-9-10-11-12-13-14-15
7	1	12	4000	1-9-10-11-12	48	6	8	1260	6-5-4-8
8	1	13	4600	1-9-10-11-12-13	49	6	9	1860	6-5-4-8-9
9	1	14	4900	1-9-10-11-12-13-14	50	6	10	2860	6-5-4-8-9-10
10	1	15	5800	1-9-10-11-12-13-14-15	51	6	11	3760	6-5-4-8-9-10-11
11	2	5	1200	2-4-5	52	6	12	4960	6-5-4-8-9-10-11-12

12	2	6	1460	2-4-5-6	53	6	13	5560	6-5-4-8-9-10-11-12-13
13	2	7	1600	2-4-5-7	54	6	14	5860	6-5-4-8-9-10-11-12-13-14
14	2	8	800	2-4-8	55	6	15	6760	6-5-4-8-9-10-1-12-13-14-15
15	2	9	1400	2-4-8-9	56	7	8	1400	7-5-4-8
16	2	10	2400	2-4-8-9-10	57	7	9	2000	7-5-4-8-9
17	2	11	3300	2-4-8-9-10-11	58	7	10	3000	7-5-4-8-9-10
18	2	12	4500	2-4-8-9-10-11-12	59	7	11	3900	7-5-4-8-9-10-11
19	2	13	5100	2-4-8-9-10-11-12-13	60	7	12	5100	7-5-4-8-9-10-11-12
20	2	14	5400	2-4-8-9-10-11-12-13-14	61	7	13	5700	7-5-4-8-9-10-11-12-13
21	2	15	6300	2-4-8-9-10-11-12-13-14-15	62	7	14	6000	7-5-4-8-9-10-11-12-13-14
22	3	6	1110	3-5-6	63	7	15	6900	7-5-4-8-9-10-1-12-13-14-15
23	3	7	1250	3-5-7	64	8	10	1500	8-9-10
24	3	8	750	3-4-8	65	8	11	2400	8-9-10-11
25	3	9	1350	3-4-8-9	66	8	12	3600	8-9-10-11-12
26	3	10	2350	3-4-8-9-10	67	8	13	4200	8-9-10-11-12-13
27	3	11	3250	3-4-8-9-10-11	68	8	14	4500	8-9-10-11-12-13-14
28	3	12	4450	3-4-8-9-10-11-12	69	8	15	5400	8-9-10-11-12-13-14-15
29	3	13	5050	3-4-8-9-10-11	70	9	12	3100	9-10-11-12

				-12-13					
30	3	14	5350	3-4-9-10-11-12-13-14	71	9	13	3700	9-10-11-12-13
31	3	15	6250	3-4-9-10-11-12-13-14-15	72	9	14	4000	9-10-11-12-13-14
32	4	6	960	4-5-6	73	9	15	4900	9-10-11-12-13-14-15
33	4	7	1100	4-5-7	74	10	12	2100	10-11-12
34	4	9	900	4-8-9	75	10	13	2700	10-11-12-13
35	4	10	1900	4-8-9-10	76	10	14	3000	10-11-12-13-14
36	4	11	2800	4-8-9-10-11	77	10	15	3900	10-11-12-13-14-15
37	4	12	4000	4-8-9-10-11-12	78	11	13	1800	11-12-13
38	4	13	4600	4-8-9-10-11-12-13	79	11	14	2100	11-12-13-14
39	4	14	4900	4-8-9-10-11-12-13-14	80	11	15	3000	11-12-13-14-14
40	4	15	5800	4-8-9-10-11-12-13-14-15	81	12	15	1800	12-13-14-15
41	5	9	1600	5-4-8-9					

Figure 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 shows the result of the Dijkstra algorithm for determining the shortest path from main gate (source node 1) to various destination nodes (buildings 4, 5, 6, 7, 10, 11, 12, 13, 14, 15).

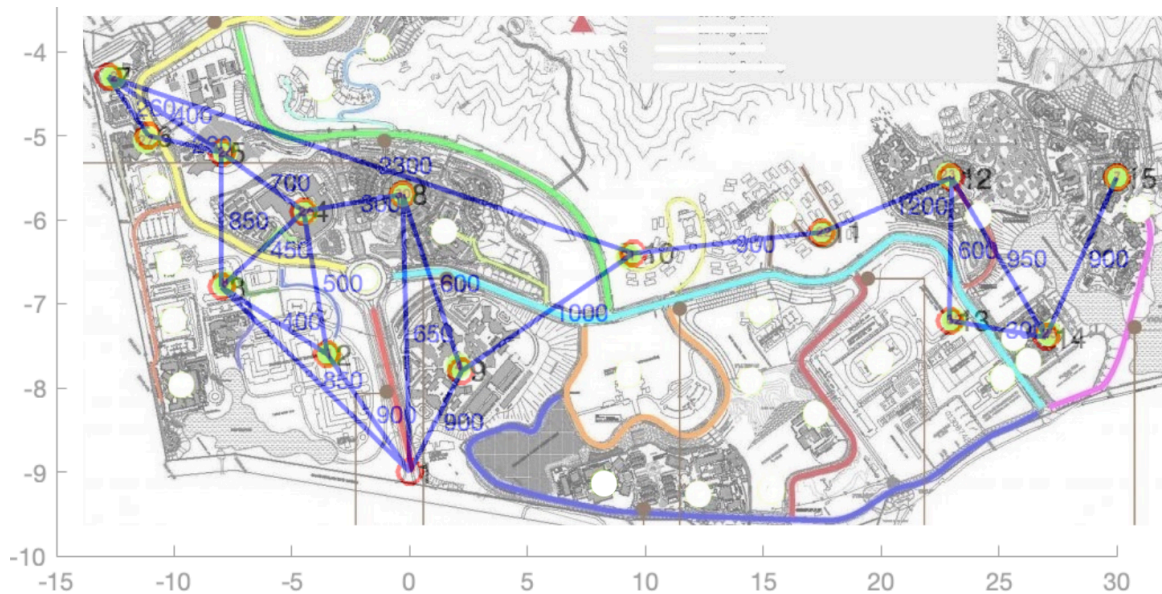


Figure 2 - The graph is on the topographic diagram of the facilities of the main campus of Universiti Malaysia Sabah

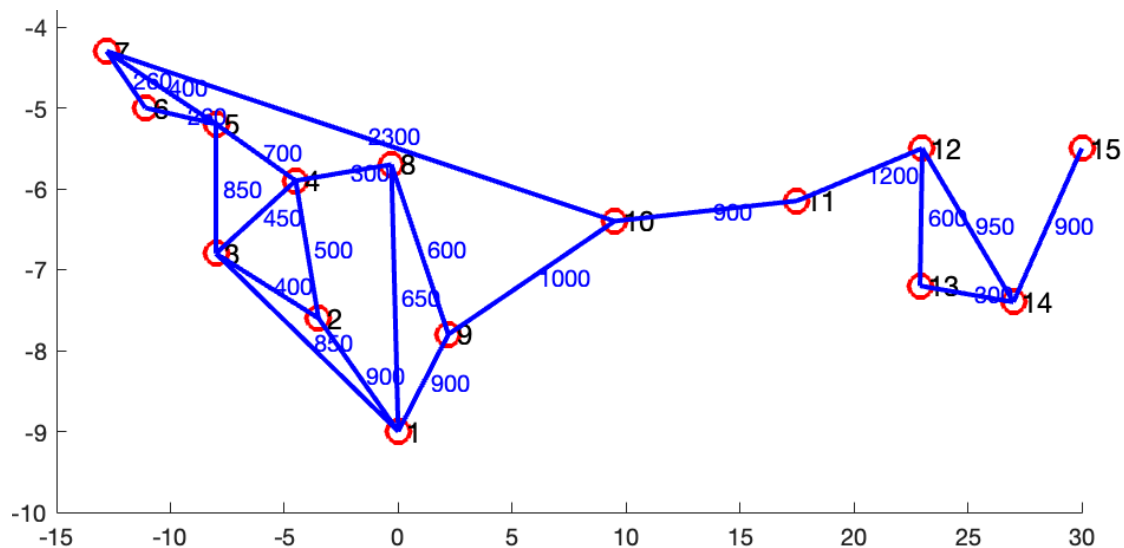
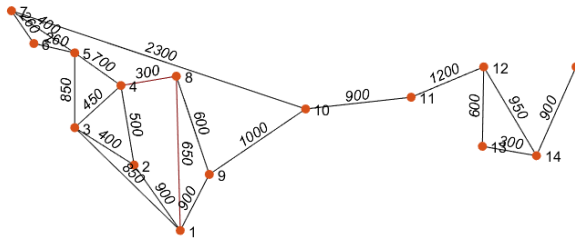
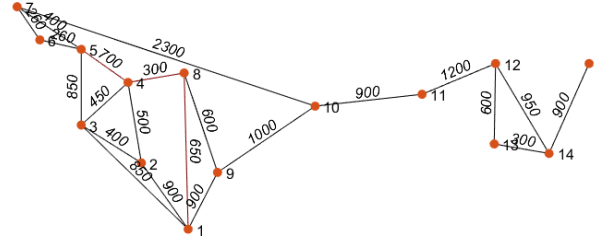


Figure 3 - The graph for the main campus of University Malaysia Sabah



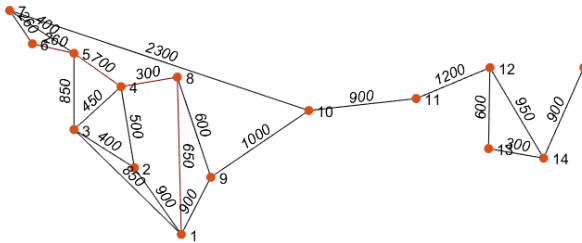
Route 1 8 4

FIGURE 4 – Outcome from the algorithms implementation - path from the main gate (node 1) to destination 4(LIBRARY)



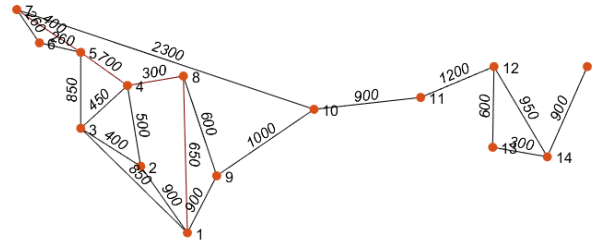
Route 1 8 4 5

FIGURE 5 – Outcome from the algorithms implementation - path from the main gate (node 1) to destination 5(DKP LAMA)



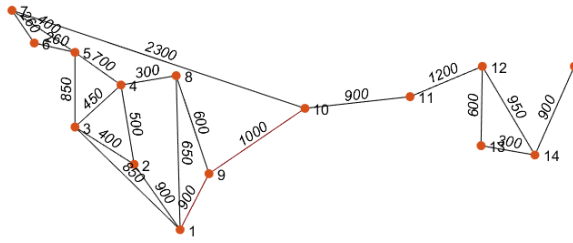
Route 1 8 4 5 6

FIGURE 6 – Outcome from the algorithms implementation - path from the main gate (node 1) to destination 6(DEWAN RESITAL)



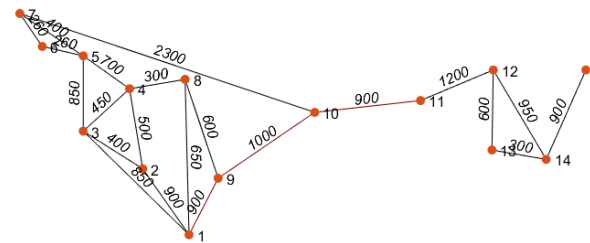
Route 1 8 4 5 7

FIGURE 7 – Outcome from the algorithms implementation - path from the main gate (node 1) to destination 7(FPEP)



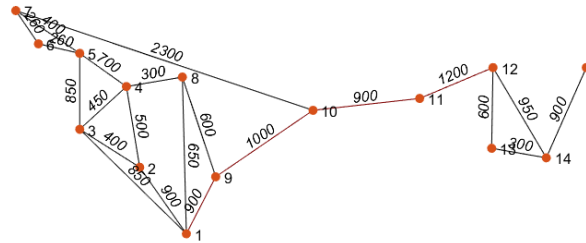
Route 1 9 10

FIGURE 8 – Outcome from the algorithms implementation - path from the main gate (node 1) to destination 10(KKTM)



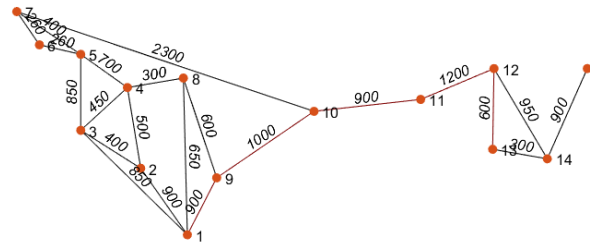
Route 1 9 10 11

FIGURE 9 – Outcome from the algorithms implementation - path from the main gate (node 1) to destination 11(KKTF)



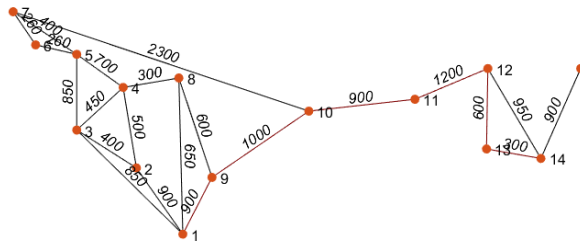
Route 1 9 10 11 12

FIGURE 10 – Outcome from the algorithms implementation - path from the main gate (node 1) to destination 12(KKTPAR)

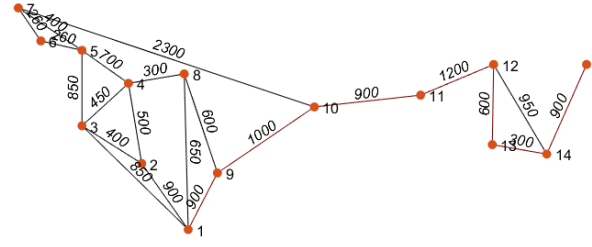


Route 1 9 10 11 12 13

FIGURE 11 – Outcome from the algorithms implementation - path from the main gate (node 1) to destination 13(PADANG KAWAD)



Route 1 9 10 11 12 13 14



Route 1 9 10 11 12 13 14 15

FIGURE 12 – Outcome from the algorithms implementation - path from the main gate (node 1) to destination 14(DKP BARU)

FIGURE 13 – Outcome from the algorithms implementation - path from the main gate (node 1) to destination 15(FPSK)

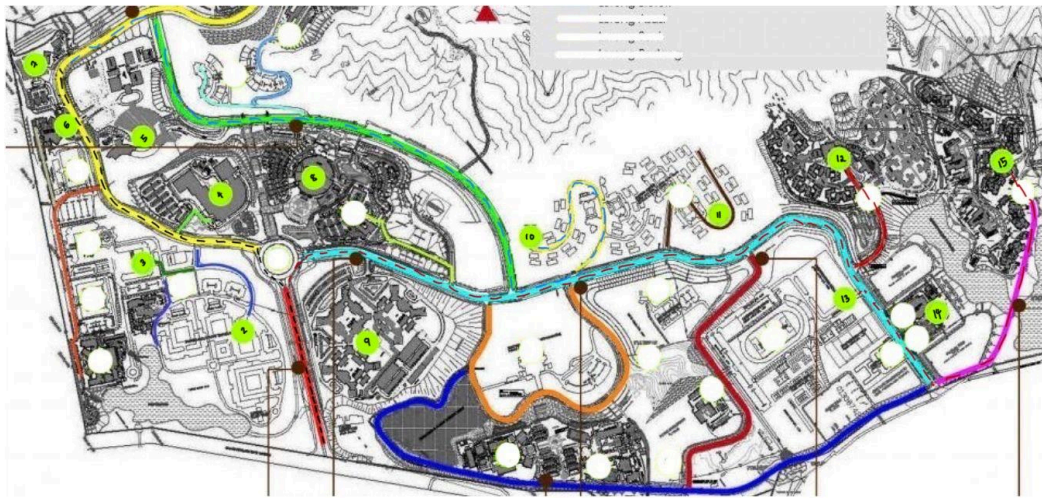


FIGURE 14 - Routes identified from the main gate (source node 1)

Table 3 shows that the longest simulated route is between nodes 6 and 15, with a distance of 6760 m and a node sequence of 6-5-4-8-9-10-11-12-13-14-15. The shortest route is between nodes 3 and 8, with a distance of 750 m and a node sequence of 3-4-8. The average route length is approximately 3285.19 m.

The mapping of specific paths in the topographic layout of campus buildings (figure 14) demonstrates the convenience of graphically showing the results.

Particular attention should be paid to the routes described from the campus's main gate (source node 1) to various buildings, as entrance into the university area is the most significant starting point for finding the various buildings and faculties. Thus, the larger routes simulated by the academic institution's main gate are: (1) 1-15, with a length of 5800 m and a sequence node of 1-9-10-11-12-13-14-15, (2) 1-14, with a length of 4900 m and a sequence node of 1-9-10-11-12-13-14, (3) 1-13, with a length of 4600 m and a sequence node of 1-9-10-11-12-13, and (4) 1-12, with a length of 4000 m and a sequence node of 1-9-10-11-12.

Given their close proximity, it is evident that routes between nodes with direct connections were not simulated; rather, the node sequence will only include the source and destination nodes. Additionally, such a calculation is not relevant for the approximation of the shortest path at the campus of the Universiti Malaysia Sabah (UMS).

In general, Dijkstra's algorithm works well for outdoor navigation, as evidenced by the UMS Main Campus, where it was successfully applied to find the shortest paths. Because of its success, it may now be used to give clear directions between departments, faculties, and other important locations within campus. The algorithm was developed and implemented correctly, as evidenced by its accurate simulation and identification of the intended paths, which guarantee dependable and effective route optimisation throughout the campus.

5.0 Conclusion

This study has aimed at implementing the Dijkstra algorithm in order to identify the shortest path between different points in the University Malaysia Sabah main campus. The Dijkstra algorithm has been applied to the campus wherein each location is represented by a node and the relationship between the locations as edges and the weights between them including the purpose of the edge. The complete process of data collection for the construction of a weighted graph representation of the UMS main campus has been a combination of mapping campus layout and the identification of important nodes, as well as determining distances or weights between these nodes. This process of using data to support decision making has been very helpful in the process of implementing the Dijkstra algorithm.

The comprehensive study provided in this document has established that the Dijkstra algorithm can be effectively used to determine the shortest routes available in the dynamic and convoluted environment of the UMS main campus. A number of tests have been conducted to determine the algorithm's effectiveness in coping with the specificities of the campus context including, the presence of buildings, the use of walkways, and potential obstacles. This is because the use of the shortest path information in campus navigation systems or signage through applications in mobile devices can greatly improve the navigation process among students, staff, and visitors on campuses. Through supplying efficient and effective routes, the campus community can reduce the amount of time spent on the journey and generally save time and energy in obtaining their desired destinations.

Apart from the domain of practical application, this research has also got academic value. Presence of a well-structured description of the used methodological approach as well as the description of the used data and the analysis of the performance of the Dijkstra algorithm can be useful as a reference for other researchers or practitioners working on further application of graph optimization techniques in navigation on campus or in the city. The conclusions of the current study can be applied to further research concerning the application of cutting-edge algorithms and technologies for better navigation and accessibility of large-scale buildings. Overall, this study has managed to identify the shortest paths across the UMS main campuses using the Dijkstra algorithm. The results of this research can help make the campus navigation experience better for the community and also explain how the graph based optimization can be applied to real world problems like this one.

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